Gabriel Schütz de Souza

Senior Software Engineer — Payments & Financial Systems

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English · German (fluent, TestDaF 5/5/5/4) · Portuguese (native) · French (B2)

Six years building and running revenue-critical payment and compliance systems in production for a German aviation SaaS: Stripe Connect pipelines, SEPA, multi-currency reconciliation, EU e-invoicing, and German tax integration — domains where errors have legal and financial consequences. Built and operated the approach-fee billing system that has invoiced and collected €62.9M, bootstrapped solo in Node.js. AI-augmented workflow; one of three core developers who kept the entire platform running. An engineer before a web developer: programming since 2014 — Assembly/C and embedded electronics first — with an internationally awarded ML + hardware assistive device built at 18.

CORE EXPERTISE

- **Payments:** Stripe Connect (destination charges, automated partner payouts, webhooks, 3D Secure), SEPA mandate lifecycle, multi-currency (EUR/CHF), refund & dispute handling, idempotent double-charge prevention
- **Billing & reconciliation:** bank-statement import with automated invoice matching (invoice number + amount + customer number), aircraft-weight-based (MTOW) approach-fee engine, dunning (Mahnwesen), periodical & recurring invoicing, audit-ready invoice sequencing
- **Regulatory compliance:** EU e-invoicing (XRechnung / EN 16931), German accounting integration (DATEV), SAP FI (RFBIBL) interface to customer production go-lives, BMD (Austria) / CEGID (France) exports, VAT & reverse-charge handling
- **AI-augmented engineering:** built *Conductor/Speccer*, a Python multi-agent orchestrator that specs, builds, and tests features autonomously (dependency-DAG scheduling, quality-gated phases); AI-driven Playwright E2E generation; agentic PR review and pre-deploy impact analysis (Claude Code in daily production use)

EXPERIENCE

Senior Software Engineer — Payments & Financial Systems

aerops GmbH (German aviation SaaS, 350+ airfields) · Jul 2020 – 2026 · Remote contract (via own consultancy)

Built and ran the platform's payments, billing, and financial-compliance layer — sole developer of its accounting-export suite (DATEV, SAP, BMD, CEGID), e-invoicing, dunning, and document dispatch; one of three core developers.

- Built and operated the **approach-fee billing system** for a national air-navigation services provider — aircraft-weight-based (MTOW) fee calculation, customer matching, invoice generation, shop integration — which has **invoiced and collected €62.9M in approach fees** (all-time; mostly bank/SEPA-settled). Bootstrapped it solo in Node.js.
- Built the German/EU **financial-compliance export suite** end-to-end: **DATEV** accounting integration, **XRechnung** e-invoicing (EN 16931, with validator), and a **SAP FI (RFBIBL) interface taken solo to two customer production go-lives** (with formal written customer sign-offs), plus **BMD** (Austria) and **CEGID** (France) exports.
- Re-architected the platform's **Stripe payment infrastructure** — inherited as a Shopware plugin — into a central payment service in the core backend: designed its event-driven spine (PaymentEvent queue + async jobs over RabbitMQ) and built Connect destination charges with automated partner payouts, SEPA mandates, 3D Secure,

EUR/CHF, refund & dispute handling, and idempotent double-charge prevention — handling **€8.47M** in gross volume across the platform.

- Built the in-house **dunning (Mahnwesen)** microservice and the **bank-statement import + payment-matching** system (camt.052 / CSV; auto-reconciles by invoice number + amount + customer number), plus periodical & recurring invoicing.
- Built a new per-airport **document-dispatch platform** (SMTP / Gmail-OAuth / Outlook-OAuth mailers with multi-tier fallback, per-mailer token-bucket rate limiting, template editor, DeepL multi-locale) replacing the legacy mailer; drove the Shopware → backend migration of the billing domain.
- Fixed request timeouts on very large orders (2,000+ line items) via a batched article-append API + async invoicing.
- Built the AI tooling behind an **AI-augmented workflow in production** on a payments/compliance codebase: *Conductor/Speccer*, a Python multi-agent orchestrator that specs, builds, and tests features autonomously, plus an AI Playwright pipeline that explores the app as a real user to derive and emit E2E test suites — and ran it where a regression carries legal and financial cost, shipping fast while keeping full architectural control.

Earlier

- **Java developer internship — SystemHaus** (Brazil, Mar–Oct 2016): first professional software role, during electronics-technician training (Assembly, C, microcontrollers)
- **Research internship — Weizmann Institute of Science** (Israel, Jul 2016): SVM-based image recognition, International Summer Science Institute
- 2017–2019: full-time computer-science studies (UNISINOS → TU Dresden) — see Education

TECHNICAL STACK

Backend: Node.js/Express, PHP/Laravel · **Frontend:** HTML, CSS, JavaScript, Blade/server-side templates (earlier projects: Vue.js, AngularJS) · **Data:** MySQL/MariaDB, MongoDB · **Payments:** Stripe (Connect, Terminal, webhooks), SEPA · **Compliance:** DATEV Connect, XRechnung · **Infra:** Docker, RabbitMQ, GitHub Actions CI/CD · **Testing:** PHPUnit, Codeception, Jest, PHPStan

EDUCATION

- **B.Sc. Computer Science — Uniritter (Brazil)**, in progress, expected 2027
- Prior CS coursework: **TU Dresden** (2019–2021); **UNISINOS** Computer Engineering (2017–2019)
- **Electronics technician** — Fundação Liberato technical school (Novo Hamburgo, Brazil), 2012–2017: Assembly, C, microcontrollers

AWARDS

ICYS Gold Medal, Engineering — Stuttgart (2017) · FEBRACE 2nd Place — São Paulo (2017) · Weizmann Institute International Summer Science Institute scholarship (2016)

The medals were for **Haptikos** (2015–16): a wearable for deafblind users — SVM facial-expression recognition on a Raspberry Pi driving a vibration-motor matrix for social-haptic communication.